LESSON PLAN

LOW SPEED FLIGHT (9)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	13	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Hydraulic system, general design requirements, typical layout of a basic hydraulic system and its brief description.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Hydraulic system, general design requirements, typical layout of a basic hydraulic system and its brief description.
5. Specific Objectives	9.1.1.1 Trainee will be able to describe various Aircraft engineering systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of hydraulic schematic diagrams; Safety: Do not operate live systems; Instructor supervision: Monitor diagram interpretation accuracy.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a hydraulic system in an aircraft?
(b) Name one major aircraft engineering system that relies on hydraulics.
(c) Why is fluid pressure preferred over mechanical linkages for actuation?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• General design requirements for aircraft hydraulics • Selection of hydraulic fluid properties • Pressure and temperature operating limits • System reliability and redundancy factors	L & Ds	PPT, IB	15 min
(b)	• Typical layout of a basic hydraulic system • Pump, reservoir, and filter arrangement • Control valves and actuator connections • Return line and accumulator integration	L & Ds	PPT, PDF	20 min
(c)	• Brief description of system operation cycle • Flow path from pump to actuator • Pressure regulation and leak management • System shutdown and isolation procedures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a hydraulic pump generates system pressure.
(b) Describe the role of the accumulator in a basic hydraulic layout.
(c) Identify two key design requirements for aircraft hydraulic fluids.

12. Home Assignment and Follow Up	Prepare a brief note comparing the typical layout of a basic hydraulic system with the alighting gear hydraulic requirements covered in LP 03.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Alighting gear, basic design requirements, types of u/c and functions.	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min